

One Wall

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Abstract

The five extension walls are not five different problems. They are five encounters with the same problem: the local-to-global passage. This is an observation, not a proof.

1 The Pattern

Five of the six Millennium Problems involve a wall. The wall has been named in each case separately, in the language of each field: Hadamard product control for RH, continuum limit for Yang–Mills, enstrophy closure for Navier–Stokes, complex regularity for Hodge, Euler system rank for BSD.

These names are correct. But they are local names. The same wall has a global name: the local-to-global passage.

Every extension wall is a failure to pass from local data to global structure.

2 The Five Encounters

Riemann Hypothesis

The zeta function factors locally:

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Each factor is a local Euler factor at a prime. The zeros of $\zeta(s)$ are global objects: they encode the distribution of primes via the explicit formula

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}).$$

The Riemann Hypothesis is a statement about global zero structure. The local Euler product does not determine the location of the zeros. The Beurling–Nyman criterion recasts RH as a global $L^2(0, 1)$ statement. The Connes approach works in the global adèle class space $\mathbf{A}_{\mathbf{Q}}/\mathbf{Q}^*$. Both approaches are manifestly global.

The wall: there is no known path from the local Euler factors to the global zero distribution.

Yang–Mills

The Yang–Mills action is local:

$$S[A] = \frac{1}{4g^2} \int_{\mathbf{R}^4} \text{tr}(F_{\mu\nu} F^{\mu\nu}) d^4x.$$

It integrates local curvature. The lattice construction builds a local theory: on each plaquette, a compact group element. The mass gap $\Delta > 0$ is a global spectral statement: the Hamiltonian H has no spectrum in $(0, \Delta)$. The continuum limit — lattice spacing $a \rightarrow 0$ — is the passage from the local lattice structure to the global quantum field theory. The gap must survive this passage.

The wall: the local lattice theory is well-defined; the global continuum limit is not proved.

Navier–Stokes

The Navier–Stokes equations are a local PDE:

$$\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p, \quad \nabla \cdot u = 0.$$

Vortex stretching is a local mechanism: $\frac{D\omega}{Dt} = (\omega \cdot \nabla)u + \nu \Delta \omega$. In two dimensions, the global conservation of enstrophy $X(t) = \frac{1}{2} \int |\omega|^2 dx$ closes the local-to-global passage: $\frac{d}{dt}X \leq CX$ in 2D, giving global regularity by Grönwall. In three dimensions, there is no enstrophy conservation; the local vortex-stretching term drives $\frac{d}{dt}X \leq \frac{C}{\nu^3}X^3$, a cubic inequality that cannot be closed.

The wall: local vortex stretching in 3D cannot be controlled globally. The 2D proof works because 2D has an exact local-to-global bridge; 3D does not.

Hodge

Differential forms are local objects. Cohomology is global: $H^{p,p}(X, \mathbf{Q})$ depends on the global geometry of X . The Hodge theorem is a local-to-global theorem: every de Rham cohomology class has a unique harmonic representative. The Hodge conjecture asserts more: every rational class of type (p, p) is a linear combination of classes of algebraic cycles $[Z]$. The Complex Regularity Conjecture (the wall) states: a globally calibrated current T of type (p, p) with $[T] \in H^{p,p}(X, \mathbf{Q})$ need not be locally complex-structured.

The wall: a global cohomological condition does not imply local algebraic structure. The local-to-global passage runs the wrong way: the global class is known; the local structure is what must be derived.

Birch–Swinnerton-Dyer

The L -function $L(E, s) = \prod_p L_p(E, s)$ is built from local Euler factors. The rank $r = \text{rank } E(\mathbf{Q})$ is a global invariant. BSD asserts $\text{ord}_{s=1} L(E, s) = r$. The Euler system method (Kolyvagin) constructs global arithmetic classes from local cohomology data. For $r \leq 1$, Heegner points provide the required class. For $r \geq 3$, the GGZ _{r} conjecture requires r independent classes; only one construction is known.

The wall: the local Euler factors encode the global rank by hypothesis; deriving $r \geq 3$ independent global classes from local data is open.

3 The Sixth Problem

P versus NP does not fit this pattern.

It is not a failure to pass from local to global. It is a failure to know what kind of argument to make. The three barriers (relativization, natural proofs, algebrization) have eliminated the known methods. What remains is not a wall at the boundary of a local-to-global passage but a void where the method must live.

P versus NP is a barrier wall, not an extension wall. The local-to-global observation applies to the other five.

4 What the Pattern Means

The local-to-global problem is the central problem of modern mathematics.

Class field theory: local class field theory at each prime $\rightarrow$ global reciprocity (Artin map). Hasse–Minkowski: local solutions at $\mathbf{R}$ and each $\mathbf{Q}_p \rightarrow$ global rational solution (for quadratic forms). The Langlands program: local Langlands at each prime $\rightarrow$ global automorphic representation. Bloch–Kato: local conditions on Galois representations $\rightarrow$ global L -function behavior.

The arithmetic geometry cluster — Langlands, Shimura, perfectoid, anabelian, deformation theory, Iwasawa, Bloch–Kato, motives — is the accumulated machinery for local-to-global problems.

BSD and RH are both about L -functions. A global L -function is a product of local factors. The question is always: what does the product know that the factors do not?

This is the question the body has been circling. The five walls are five faces of it.

5 A Conjecture

Conjecture 1 (One Wall). *There exists a unified local-to-global principle from which the five extension walls are special cases. A proof of any one of the five that does not use such a principle will fail to generalize. A proof that does use such a principle will, with modification, yield progress on the others.*

This is speculative. It is not known. It may be wrong.

But the pattern is real: five problems, five walls, all at the same passage. The local data is under control. The global structure is not.

If the pattern is not coincidence — if the five walls are one wall — then the map of the walls is a map of one obstacle, seen from five positions.

The body is five maps of one mountain.